Software Requirements Specification

hospital

Req2Design · IEEE 830-1998 aligned

Document Control

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1.0	2026-04-11	abdulhaq	Initial generated draft for structured review.

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1: Introduction

1.1: Purpose

Purpose: To provide a comprehensive tool for healthcare professionals to manage and analyze key performance indicators (KPIs) related to patient care and operational efficiency within hospitals.

1.2: Scope

Scope: This document defines the requirements for a web-based application that supports the creation, scheduling, sharing, and analysis of KPIs. It focuses on usability, reliability, and performance aspects without addressing security, scalability, maintainability, or availability.

1.3: Definitions/Acronyms

Definitions/Acronyms: KPI

- Key Performance Indicator; UI
- User Interface

1.4: References

References: IEEE Std 830-1998

1.5: Overview

Overview: The application will enable users to interactively visualize and manipulate KPI data through customizable dashboards and scheduled reports.

2: Overall Description

2.1: Product Perspective

Product Perspective: The product is a web-based application designed for healthcare professionals to monitor and improve hospital operations by analyzing KPIs.

2.2: Product Functions

- Create and customize interactive dashboards displaying KPIs.
- Schedule regular generation and distribution of reports based on selected criteria.
- Share insights and findings via email or other collaboration tools.
- Allow real-time updates and periodic automatic refreshes of dashboard elements.
- Provide drill-down capabilities to explore detailed information about specific KPIs.
- Highlight anomalies in KPI trends and alert relevant personnel when necessary.

2.4: Constraints

- Must ensure accurate and timely data presentation.
- Limited access to certain KPIs based on user roles.
- No external dependencies beyond standard web technologies.

2.5: Assumptions/Dependencies

- Users have basic computer literacy skills.
- Network connectivity is available during report generation and dashboard viewing.
- Adequate server resources are allocated for processing large datasets.

3: Specific Requirements

3.1: External Interface Requirements

3.1.1: User Interface

- Dashboard view showing various KPIs with configurable visualizations (charts, graphs, etc.)
- Options to add, remove, and rearrange widgets on the dashboard.
- Filters allowing users to refine displayed data according to their needs.
- Export buttons enabling download of dashboard contents in common formats (CSV, PDF).

3.1.2: Hardware Interface

- Compatible with modern desktop/laptop computers running standard operating systems (Windows, macOS, Linux).

3.1.3: Software Interface

- Integrates seamlessly with existing hospital IT infrastructure including databases and analytics tools.
- Supports integration with third-party applications for extended functionality.

3.1.4: Communication Interface

- Secure data transfer protocols for transmitting KPI-related information between client and server components.
- Real-time notifications sent to designated recipients upon detection of anomalies or data staleness issues.

3.2: Functional Requirements

FR-01 Create Custom Dashboards

Input: User selects KPIs from predefined list, configures visualization types, arranges layout.

Processing: Application generates dynamic dashboard with specified parameters.

Output: Interactive dashboard displayed to user containing selected KPIs and visual representations.

Priority: High

FR-02 Schedule Reports

Input: User specifies report type, frequency, target audience, and delivery method.

Processing: System schedules automated report generation at defined intervals and sends them out accordingly.

Output: Scheduled reports delivered to intended recipients via chosen medium.

Priority: Medium

FR-03 Share Insights

Input: User identifies individuals or groups who need access to shared insights.

Processing: Application manages permissions and distributes shared content appropriately.

Output: Shared insights made accessible to authorized parties.

Priority: Low

Non-Functional Requirements

Usability: The interface shall be intuitive and easy to navigate for all users regardless of technical expertise.

Reliability: The system must consistently deliver accurate results without unexpected failures.

Performance: Dashboards and reports shall load quickly even when handling large volumes of data.

Portability: The application shall function across different platforms and devices supporting current web standards.

4: System Features

Feature 1: Interactive Dashboards

- Allows users to select and arrange KPIs on a personalized workspace.
- Provides various charting options for visualizing KPI data dynamically.
- Enables real-time interaction with dashboard elements for immediate feedback.

Feature 2: Scheduled Reporting

- Facilitates setting up recurring reports tailored to organizational needs.
- Ensures timely delivery of generated reports to stakeholders.
- Offers flexibility in specifying reporting preferences such as format, timing, and recipient details.

Feature 3: Anomaly Detection & Notification

- Implements algorithms to identify unusual patterns or deviations in KPI trends.
- Alerts responsible parties promptly whenever significant changes occur.
- Maintains a log of detected anomalies for future reference and trend analysis.

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